

LESSON 8.1

AREA OF TRIANGLES – PART 1



LESSON INTRODUCTION

MA.6.GR.2.1 Derive a formula for the area of a right triangle using a rectangle. Apply a formula to find the area of a triangle.

LEARNING TARGETS

- I can discover the formula for the area of a triangle.

BENCHMARK COHERENCE

BUILDING ON

MA.5.GR.2.1 Find the perimeter and area of a rectangle with fractional or decimal side lengths using visual models and formulas.

WORKING TOWARD

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

ESSENTIAL QUESTIONS

- What is the formula to find the area of a triangle?
- How can you use the area of a rectangle to find the area of a triangle?

LESSON NARRATIVE

In the lesson, students will discover the formula for the area of a triangle by identifying how the area of a triangle relates to the area of a rectangle. During each exploration of a right triangle, acute triangle, and obtuse triangle, students will discover that the area of a triangle is always half the area of a rectangle with the same base and height. Students apply this knowledge to determine the formula to find the area of a triangle.

COMPONENT SUMMARY

8.1.1 WARM-UP (~5 MIN.)

Open-ended response on the importance of struggling

8.1.3 COLLABORATION (20-25 MIN.)

Use the area of a rectangle to find the area of an obtuse and acute triangle

8.1.5 WRAP-UP (~5 MIN.)

Students analyze their own understanding of how to find the area of a triangle

8.1.2 EXPLORATION (10-15 MIN.)

Use the area of a rectangle to discover how to find the area of a triangle

8.1.4 BRING IT TOGETHER (5 MIN.)

Students synthesize information connecting area of a rectangle to a square

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Base
- Height
- Acute
- Obtuse

MATERIALS

- Importance of Struggle video
- Paper triangles
- Scissors
- Glue (or tape)
- (Optional) Graph paper

ADDITIONAL PREPARATION

- 8.1.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- In finding the height of a triangle, students may pick a side of an acute or obtuse triangle instead of the height.
- Students might forget to multiply the length and width by $\frac{1}{2}$ to find the area of a triangle.
- Students may forget to find the height of an obtuse triangle by extending the line.
- In acute triangles, students may incorrectly try to extend the base to the dashed height line.

THINGS TO CONSIDER

- The 8.1.2 Exploration is where students will be able to visualize how the formula for area of a triangle works and relates to a rectangle.
- The lesson is a discovery of the formula for the area of a triangle. The formula is revealed in the lesson. Allow students to work through the discovery to determine the formula.
- The visual in 8.1.3 Collaboration question 5 will help show students how the area of parallelograms can also relate to the area of a triangle and what to do with triangles that do not have right angles.

LITERACY AND LANGUAGE ROUTINES

Number Talks

Consider using a Number Talks strategy discussion to visualize different ways to divide a rectangle. Draw a rectangle on the board. Ask students to come up with different ways to divide the rectangle in half. Have students share strategies until someone splits the rectangle diagonally.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

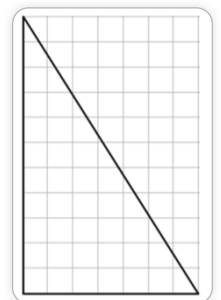
MA.K12.MTR.2.1 Multiple Representations

Throughout this lesson, students have the opportunity to make connections between the areas of various shapes, such as rectangles and parallelograms, and the area of triangles. Encourage students to express these connections through discussion as each activity allows.

DIFFERENTIATE INSTRUCTION

Consider opportunities for students to engage in physical or digital manipulatives that offer kinesthetic entry points for making sense of the formula for the area of a triangle through the lens of a rectangle.

- Coordinate Plane Cutouts: Students can compose and/or decompose triangles and rectangles using mini coordinate planes to highlight the relationship between their area formulas.
- Tangram Triangles: Similarly, students can engage with the digital tangram triangles that target the relationship between rectangles and triangles and what that means for their area.



8.1.1 WARM-UP: THE IMPORTANCE OF STRUGGLE

Component Overview: Students will watch a video on the importance of struggling. Students will then answer open-ended questions about how they have personally struggled.

TEACHER MOVES

Consider positioning students to discuss when they have been in a situation that contained each option to engage auditory learners and put real-life context to the options. Take the discussion further and have students turn and talk about why failing is sometimes okay. Share a time you as an adult have failed and how it made you stronger.



8.1.1 Warm-Up: The Importance of Struggle

When learning something new, whether it is a sport, a musical instrument, or even math, you may experience failure before you experience success. The key is to keep going even when you make mistakes. Learning takes place during the time you practice and during the time you struggle.

1. Which choice would you feel less comfortable doing?

- Answering a question and getting it wrong.
- Saying, "I do not know," as your answer to a question.

Answers will vary.

2. Describe a time that you persevered when learning something new.

Answers will vary.

3. When you persevered in learning, how did you overcome any initial fear(s)?

Answers will vary.



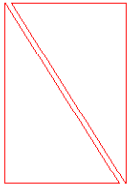
8.1.2 Exploration: Discovering the Area of a Right Triangle

A **right triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.



- 1. Trace two copies of the right triangle on paper, then cut them out.
- 2. Form a rectangle using the two right triangles. Paste the triangles here.
- 3. Using the image just created, complete the table.

	Rectangle	Triangle
Height	11 cm	11 cm
Base	7 cm	7 cm



- 4. What is the area of the rectangle? Include units.
 77 cm^2
- 5. Describe the relationship between the area of one right triangle and the area of the rectangle.
The area of a triangle is one-half the area of the rectangle.
- 6. What is the formula for the area of a rectangle?
 $A = bh$
- 7. Use the relationship observed and the formula for the area of a rectangle to write a formula that can be used to find the area of a right triangle.
 $A = \frac{1}{2}bh$

8.1.2 EXPLORATION: DISCOVERING THE AREA OF A RIGHT TRIANGLE

Component Overview: In this Exploration, students discover how the area of a rectangle is connected to the area of a right triangle. Begin with having students trace and cut out triangles. They will use these triangles to create a rectangle. The questions in this section all pertain to one set of triangles. Spend time exploring the connection between right triangles and rectangles, as the following sections will give more opportunity for actual calculations.

TEACHER MOVES

Number Talks

Consider using a Number Talks strategy discussion to visualize different ways to divide a rectangle. Draw a rectangle on the board. Ask students to come up with different ways to divide the rectangle in half. Have students share strategies until someone splits the rectangle diagonally.

DIFFERENTIATE INSTRUCTION

Cutting and pasting will allow an entry point for kinesthetic learners and allow students to visualize how two triangles can create a rectangle.

ENRICHMENT

- To encourage students to make the connection between rectangles and triangles, ask questions like:
- What is the difference between a regular triangle and a right triangle?
 - What does a rectangle have in common with a right triangle?
 - What do you notice about the height and base of the triangle compared to the rectangle?
 - What two types of measurement do you need to find the area of a triangle? What two types of measurement do you need to find the area of a rectangle? What is a different way you could write the area of a triangle?

8.1.3 COLLABORATION: AREA OF AN ACUTE AND AN OBTUSE TRIANGLE

Component Overview: In this Collaboration, students will work with a partner to find areas involving acute and obtuse triangles. Students then work with a partner to calculate the area of a triangle inscribed in a rectangle along with the rectangle and to find the area of obtuse triangles.

DIFFERENTIATE INSTRUCTION

On question 1, have students label all measurements they were given to visualize the connection as well as digest the information. Students may also need to draw the line for the height in Diagram 1 in the middle of the triangle to help visualize. This provides visual entry points into discussion through the diagrams.

TEACHER MOVES

Avoid student focus on the other triangles that are made outside of the triangle using the corners of the rectangle, as students may still be looking for right triangles. Discuss how a right angle should be created when finding the height of an acute or obtuse triangle.

Relating the height of Triangle A to the height of a mountain may help students understand that the height is the straight line distance from the ground (the base) to the top of the mountain (the vertex).

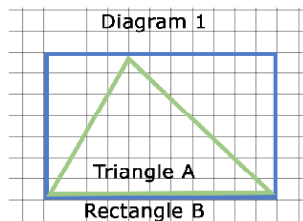
TEACHER MOVES

Consider briefly reviewing or providing definitions for acute and obtuse triangles.

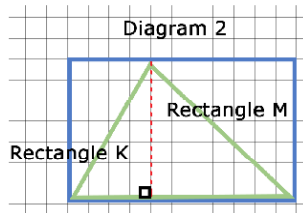


8.1.3 Collaboration: Area of an Acute and an Obtuse Triangle

An **acute triangle** is shown. Each square of the grid paper in Diagram 1 and Diagram 2 is 1 centimeter (cm) by 1 cm.



Rectangle B is drawn around acute Triangle A, as shown.



The height of acute Triangle A is shown as a dashed red line. It divides Rectangle B into two smaller rectangles, Rectangle K and Rectangle M, and divides Triangle A into two smaller right triangles.

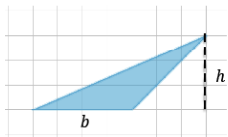
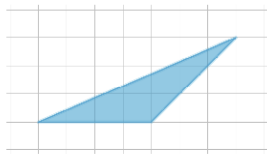
1. Use the two diagrams to complete the table.

	Area of the Rectangle	Area of the Right Triangle Drawn Inside the Rectangle
Rectangle K	28 cm^2	14 cm^2
Rectangle M	49 cm^2	24.5 cm^2

2. The area of acute Triangle A is equal to the sum of the right triangle inscribed in Rectangle K and the right triangle inscribed in Rectangle M.

- What is the area of acute Triangle A?
 38.5 cm^2
- What is the area of Rectangle B?
 77 cm^2
- What is the relationship between the area of Rectangle B and the area of acute Triangle A?
If you double the area of Triangle A, it is the same area as Rectangle B.

3. An **obtuse triangle** is shown. Each square of the grid paper is 1 centimeter (cm) by 1 cm.



- What is the length of the triangle's base (b)?
 4 cm
- What is the height (h) of the triangle?
 3 cm

8.1.3 COLLABORATION: AREA OF AN ACUTE AND AN OBTUSE TRIANGLE (CONTINUED)

DIFFERENTIATE INSTRUCTION

8.1.3 Collaboration implements visual entry points into discussion using the diagrams. Cutting and pasting the triangles into rectangles will help kinesthetic learners relate a triangle with being half of a rectangle. Consider a Turn and Talk if some students dominate more of the conversation or if your students benefit from structured timing for conversations.

ENRICHMENT

Students may struggle to find the height of triangles that are not right triangles because it may not be as clear visually. Relating the obtuse triangle to the idea of a mountain may help students visualize. Or ask students what real-world object an obtuse triangle reminds them of and use that to query about height of the triangle from the ground.

Ask prompting questions to guide students toward determining how they can find the height:

- Why isn't the longest side of the triangle the height?
- How do you know how far to draw the dotted line for the height of an obtuse triangle?

QUESTIONING

Reinforce the concept that the height needs to be perpendicular to the base. Extend the discussion by asking students to draw a different line to find the height.

8.1.3 COLLABORATION: AREA OF AN ACUTE AND AN OBTUSE TRIANGLE (CONTINUED)

ENRICHMENT

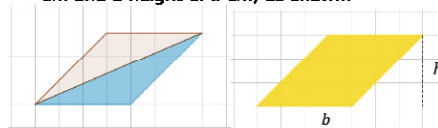
Use inquiry-based learning to help students draw connections between parallelograms, rectangles, and triangles. This will build conceptual understanding as students reason about how to find the area of a triangle.

- What do you know about parallelograms?
- What is the area of the parallelogram?
- How do the diagrams show relationship between the area of a parallelogram and the area of a rectangle?
- How is the height of a parallelogram related to the height of an obtuse triangle?

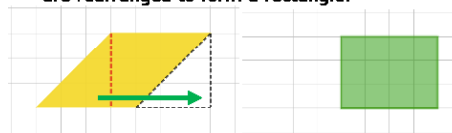
DIFFERENTIATE INSTRUCTION

Consider using manipulatives, such as tangrams, to rearrange the parallelogram into a rectangle. This will help students who have trouble with spatial reasoning. What do you notice about the height and base for the rectangle and triangle? What do you wonder about their areas?

4. A copy of the triangle is made, and the two triangles are arranged to form a parallelogram with a base of 4 cm and a height of 3 cm, as shown.



The parallelogram is then decomposed, and the parts are rearranged to form a rectangle.



- a. Complete the table.

	Rectangle	Obtuse Triangle
Height	<u>3</u> cm	<u>3</u> cm
Base	<u>4</u> cm	<u>4</u> cm

- b. How many obtuse triangles were needed to create the rectangle?

2

- c. What is the area of the rectangle?

12 cm²

- d. What is the relationship between the area of the rectangle and the obtuse triangle?

Answers will vary. The area of the rectangle is double the area of the obtuse triangle. The area of the obtuse triangle is half the rectangle.

- e. Based on this relationship, what is the area of the obtuse triangle?

6 cm²



8.1.4 Bring It Together: Formula for the Area of a Triangle

1. Complete the statements.

The area of the **right** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

The area of the **acute** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

The area of the **obtuse** triangle was ☐ half ☐ the same as the area of the rectangle with the same base and height.

2. The formula for the area of any rectangle is $A = bh$.

$$A = \text{base} \cdot \text{height}$$

The formula for the area of any triangle is $A = \frac{1}{2}bh$.

$$A = \frac{1}{2}(\text{base} \cdot \text{height})$$

8.1.4 BRING IT TOGETHER: FORMULA FOR THE AREA OF A TRIANGLE

Component Overview: Students synthesize the information they learned in the 8.1.2 Exploration and 8.1.3 Collaboration to create concise definitions and formulas for the area of a right, acute, and obtuse triangle.

TEACHER MOVES

Use this section to encourage synthesis, not memorization. Consider using this section as notetaking for students during the 8.1.3 Collaboration activity. Students may need to draw a small diagram next to each description to help visualize their thinking.

DIFFERENTIATE INSTRUCTION

Alternatively, if time permits, prompt students to write a note to an absent student, a sibling, or a family member explaining how they can use rectangles to find the area of triangles.



8.1.5 Wrap-Up: Reflection

1. Place an **X** on the line to indicate your mastery of today's lesson.

I can explain why the area of a triangle is half the area of a rectangle with the same base and height for right triangles, acute triangles, and obtuse triangles.

I need help. I'm getting there. I understand and have evidence.

←-----→

2. Write a brief explanation of your placement of the X.

Answers will vary.

8.1.5 WRAP-UP: REFLECTION

Component Overview: In this Wrap-Up, students will reflect on their understanding of the lesson. Students will place an X on the line to visually describe their understanding of the lesson. Students will explain their placement to self-reflect on what they know and what they need to improve on.

DIFFERENTIATE INSTRUCTION

For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example.